Assignment 5: ViT-S LoRA & Adversarial Attacks

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Repository and Project Links

- **GitHub Branch:** Link to Assignment 5 Branch
- **WandB Project:** Link to Weights & Biases Dashboard
- **HuggingFace Model:** Link to Best Q1 Model

1 Q1: ViT-S Fine-Tuning and LoRA on CIFAR-100

1.1 Training Logs and Metrics

Experiment 1: Rank: 2, Alpha: 2, Dropout: 0.1

Epoch	Training Loss	Validation Loss	Training Accuracy	Validation Accuracy
1	1.2159	0.4672	72.51%	86.57%
2	0.3830	0.3861	88.43%	88.16%
3	0.3202	0.3617	89.99%	88.84%
4	0.2802	0.3483	91.18%	89.39%
5	0.2543	0.3354	92.07%	89.58%
6	0.2350	0.3292	92.50%	89.68%
7	0.2182	0.3245	93.16%	89.94%
8	0.2091	0.3231	93.40%	90.03%
9	0.2017	0.3210	93.74%	90.11%
10	0.1991	0.3210	93.77%	90.08%

Table 1: *Training metrics for Rank 2, Alpha 2*

1.2 Final Testing and Hyperparameter Comparison

LoRA Layers	Rank	Alpha	Dropout	Overall Test Acc	Trainable Params
Without	-	-	-	80.87%	38,500
With	2	2	0.1	90.11%	93,796
With	2	4	0.1	89.80%	93,796
With	2	8	0.1	89.92%	93,796
With	4	2	0.1	90.09%	149,092
With	4	4	0.1	89.97%	149,092
With	4	8	0.1	90.20%	149,092
With	8	2	0.1	89.96%	259,684
With	8	4	0.1	90.09%	259,684
With	8	8	0.1	90.37%	259,684
Best (Optuna)	**?**	**?**	**?**	**% (Best)**	**?**

Table 2: *Overall Comparison of LoRA Configurations*

1.3 Graphs and Visualizations (Q1)

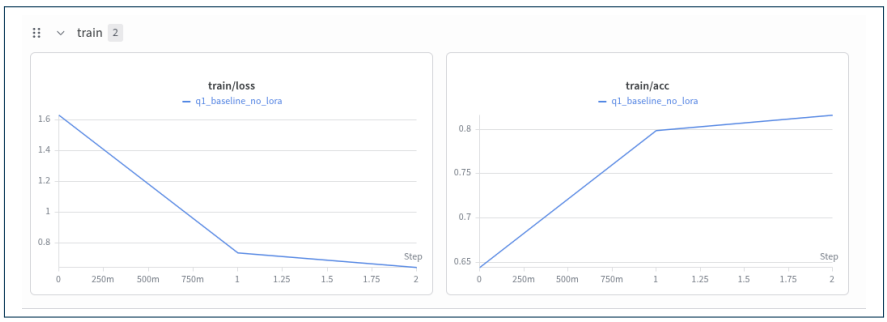

Figure 1: **Train-Val Loss/Accuracy Graphs for the Best Model**

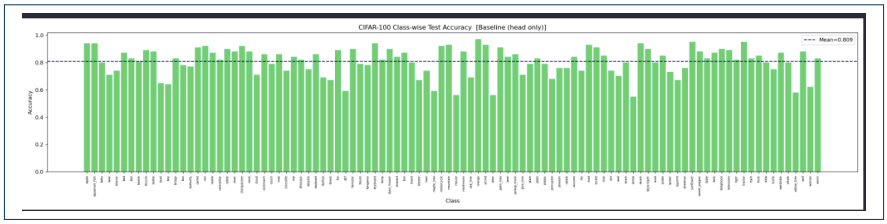

Figure 2: **Class-wise Test Accuracy Histogram**

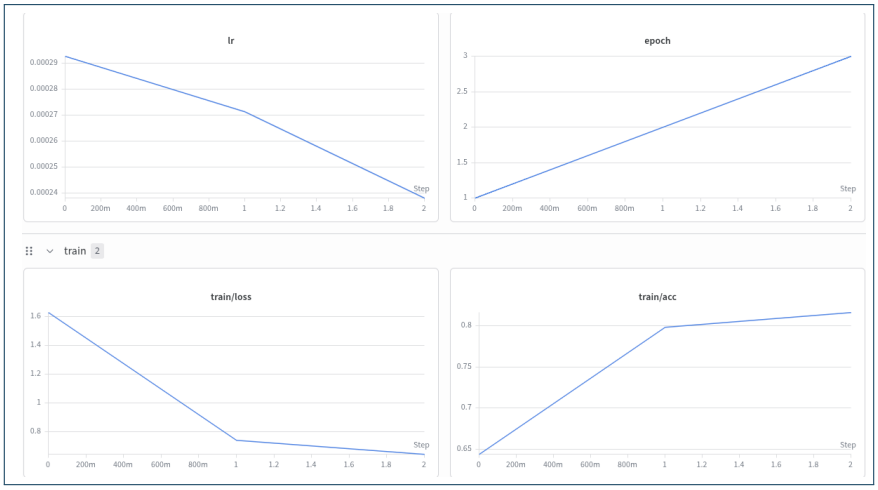

Figure 3: **Gradient Updates on LoRA Weights during Training (From WandB)**

2 Q2: Adversarial Attacks (IBM ART) on ResNet-18

2.1 Fast Gradient Sign Method (FGSM)

ϵ (Epsilon)	Clean Accuracy	Adversarial Acc (Scratch)	Adversarial Acc (ART)
0.01	89.50%	70.00%	76.00%
0.03	89.50%	34.10%	40.70%
0.05	89.50%	18.70%	25.40%
0.10	89.50%	8.10%	14.30%

Table 3: *Performance drop against Perturbation Strength*

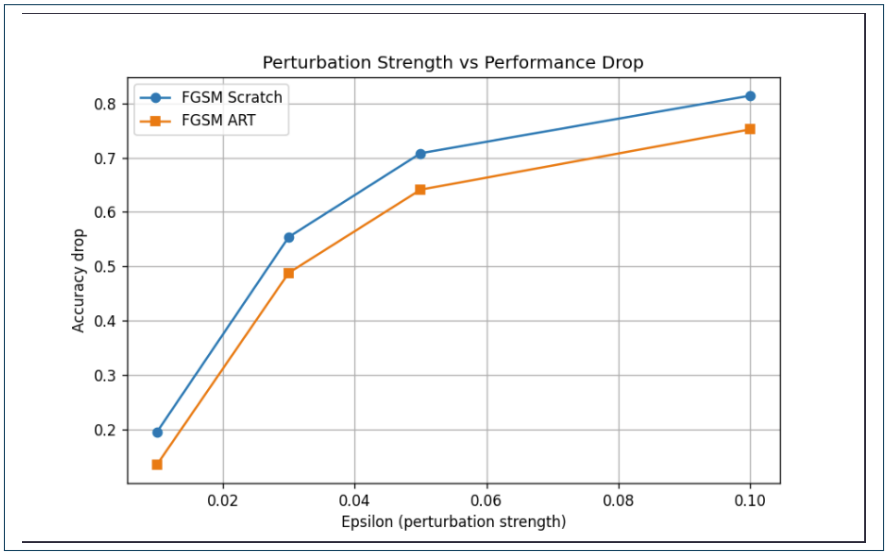

Figure 4: **Accuracy vs. ϵ Perturbation Strength**

Visual Comparison of Adversarial Images

Figure 5: **Qualitative Samples: Original vs. FGSM Scratch vs. FGSM ART**

2.2 Adversarial Detectors (ResNet-34 Binary Classification)

Target Attack Method	Binary Detection Accuracy
PGD Attack Detection	50.30%
BIM Attack Detection	50.05%

Table 4: *Performance of the ResNet-34 Detectors on PGD and BIM samples*

Analysis: PGD vs BIM Detection Difficulty

Insert your brief analysis here comparing the detection difficulties of PGD versus BIM attacks based on your accuracy observations...